1. Current State of the European Space Strategy

The space industry is changing a lot right now. In the past, only big governments explored space. Today, it is a global market filled with private companies. The space market is currently worth about $630 billion. Experts believe it will grow to $1.8 trillion by 2035. In this fast-moving world, the European Union (EU) faces big challenges. Problems in the economy and glob al politics show that Europe has weak spots. These weak spots are in its supply chains, its ability to launch rockets, and its technology independence.

1.1. The Macroeconomic and Geopolitical Landscape

Europe's space plan is deeply affected by the economy and global events. The 2025 OECD Economic Survey says Europe's economy is recovering slowly. This is mainly because of the war in Ukraine and trade tensions around the world. Because of this, European leaders had to rethink their budgets. They must save money, but they also need to spend a lot on defense and energy. As a result, space is now seen as both a business tool and a military tool. It is vital for Europe's wealth and safety.

In June 2025, the European Commission shared its "Vision for the European Space Economy." This is a new plan to make the EU a global space leader by 2050. It is a big shift. For the first time, Europe is looking at the whole space system. This includes factories, operations in space, and new ideas like mining in space.

This vision is part of the new EU Space Act of 2025. Before, European countries had 13 different sets of rules for space. This new law creates one single set of rules for everyone. It makes strong rules for space traffic, cyber safety, and protecting the environment. These rules apply to any company working in the EU, even foreign ones.

The European Space Agency (ESA) got a 30% budget increase in late 2025, reaching €22.1 billion. But even with this money and new laws, Europe has a big crisis right now. Europe cannot easily launch its own rockets. They can no longer use Russian Soyuz rockets. Also, the old Ariane 5 rocket is retired, and the new Ariane 6 is delayed. This leaves Europe very vulnerable.

In 2024, Europe only launched 3 rockets. In contrast, the US launched over 150, and China launched 70. Because of this, Europe had to pay foreign companies, like SpaceX, to launch important European satellites (like the Galileo system). This shows that Europe urgently needs a real plan to connect its big policy ideas with actual manufacturing.

1.2. Comparative Analysis: The Strategic Philosophies of France and Germany

The European Union wants a single space plan for everyone. However, the real power lies in the domestic rules of its biggest countries. France and Germany are the two main engines of Europe's space industry. But they have very different ideas on how to succeed. These differences come from their unique histories and industries.

On one side, France believes the government should lead the way. Their main goals are national security and keeping control of their own space tools. The French space agency (CNES) strongly supports the big Ariane rockets. It is also the biggest buyer for many military space

projects.

France treats space like a military zone. Paris even created a special Space Command. They spend a lot of money on "patrol" satellites and sensors to protect their space equipment from enemies. Also, France has a huge geographic advantage. It hosts Europe's main spaceport in Kourou, French Guiana. This spaceport is absolutely necessary for Europe to launch rockets on its own.

The French government keeps putting money into this top-down system. Recently, they added €500 million to the CNES budget through the "France 2030" plan. This money will help start new projects in telecommunications and future rockets. In short, the state controls the industry and sees space as a way to show national strength.

On the other hand, Germany focuses on business and the "New Space" idea. They want a diverse market with many different companies. In 2025, Germany adopted its first official space strategy. Berlin promised to spend about €35 billion over five years. This shows they want to invest more money and have a louder voice in Europe's space plans.

The German space agency (DLR) does not just rely on old, giant aerospace companies. Instead, they strongly support new, private start-ups that build small rockets. Companies like Isar Aerospace and Rocket Factory Augsburg are great examples. Germany wants to capture the fast-growing market for small commercial rocket launches.

In terms of technology, Germany is leading research on reusable rockets to make launches cheaper. They are running projects like CALLISTO (a joint project with Japan and France), ReFEx, and the LUMEN rocket engine.

These two different styles (the French government model and the German business model) create a mix of strength and tension. Having different ideas makes Europe more resilient, but it also makes things complicated.

2. Supply Chain Self-Sufficiency: The Chinese Paradigm

To plan the future of Europe's space supply chain, we must look at what rival countries are doing. China wants total independence in its space and tech industries. Chinese leaders believe that relying on foreign technology, especially from the US or Europe, is a huge weakness. It leaves them open to being blocked or sanctioned by other countries. Because of this, they see international mixing as a risk, not a benefit. This goal of complete independence is very clear when we look at China's space station and the computer chips that run it.

2.1. The Tiangong Space Station: Independent Operations

The Chinese Space Station, named Tiangong (meaning "Heavenly Palace"), is the best example of China's independent space system. It orbits between 340 and 450 kilometers above Earth and weighs about 100,000 kilograms. It has three main parts: the Tianhe core module, and the Wentian and Mengtian laboratory modules.

Building the International Space Station (ISS) was hard because astronauts had to do spacewalks to connect cables and pipes by hand. Tiangong is different. Its modules were fully built and ready to go before they were launched. This shows China has a very fast and efficient manufacturing process. Also, the ISS relies on many countries working together to send supplies and astronauts. This can be tricky when global politics get tense. In contrast, China takes care of Tiangong all by itself.

China keeps its station running using advanced robots and smart logistics. Tiangong has several robotic arms that do the dangerous work outside, so astronauts do not have to do as many spacewalks. For example, the Tianhe module has a large 10-meter (33-foot) arm, and the Wentian module has a 5-meter (16.4-foot) arm. Both arms can crawl around the outside of the station to move heavy modules and catch visiting spacecraft. Inside the Mengtian module, there is a small, very precise robotic arm. It is about 70 to 80 centimeters long. It is incredibly accurate (down to 0.1 millimeters), helping astronauts with delicate science experiments.

Along with these smart robots, China has a very fast, independent system to launch rockets and supplies. A great example of this happened when space debris damaged the Shenzhou-20 spacecraft. China's space agency prepared and launched an empty rescue ship (Shenzhou-22) in just 16 days. This incredible speed shows the strength of their system. By using their own Tianzhou cargo ships and Shenzhou crew ships, China can safely keep a permanent human presence in space. They do not need to ask other countries for help, even during an emergency.

2.2. The Semiconductor Chokepoint: China’s Quest for Radiation-Hardened Autonomy

To truly measure China's independence in space, we must look at very small things, not just giant space stations. We need to look at electronic components. Space missions require highly specialized computer chips. These are called "radiation-hardened" chips. They must be tough enough to survive extreme heat, cold, the vacuum of space, and dangerous cosmic radiation.

China has a big strategy called "Made in China 2025." At first, their goal was to make 50% of their own computer chips by 2020. Officially, they missed this goal by a lot, only reaching 16.6%. However, we must be very careful with these numbers. In the defense and space sectors, countries often hide their true strength. Different experts disagree on China's success. Some US reports highlight that China failed to meet its target. But other experts, like Bloomberg, say China is actually very strong and competitive in making specialized, older-style chips.

Because the US put strict limits on what technology China can buy, Beijing is spending huge amounts of money. In 2024, they launched a massive $49 billion fund for their chip industry. The government also has a strict, unwritten rule. When Chinese chip factories expand, they must use at least 50% Chinese-made equipment. Their ultimate goal is to use 100% local equipment.

In the space sector, groups tied to the military lead this push for independence. A key player is the Beijing Microelectronics Technology Institute (BMTI). They act as the microelectronics research branch for the Chinese military. BMTI makes highly reliable chips, processors, and communication parts designed specifically for the harsh environment of space.

For example, BMTI manufactures exact copies of Western chips (like the American AD9361 part). Chinese defense engineers are required to use local factories, like SMIC, to build these components. Because of this closed loop, all of China's space electronics, from sensors to main computers, are completely protected from foreign sanctions or supply chain problems.

2.3. Quantitative Comparative Analysis: European vs. Chinese Autonomy

People talk a lot about super advanced computer chips, like 3nm or 5nm sizes. But for space, military, and factories, older and highly reliable chips are much more important. These are called "legacy" chips, meaning they are 28nm or larger. This is exactly where Europe is very weak compared to China's growing power.

Right now, China controls about 30% of the global market for these legacy chips. They are also planning to build 40% of all the world's new legacy chip factories by 2030. On the other hand, the European Union (EU) only holds 13% of this important market.

By 2030, the EU will need about 12.7 million wafers of legacy chips every year. But Europe's planned factories will only make about 4.5 million wafers a year. This creates a huge gap. Europe will have to import at least 8.2 million wafers every year. To fix just 80% of this shortage locally, the EU would need to build 21 new chip factories. This would cost a massive amount of money and is very unlikely to happen anytime soon.

Also, the types of factories being built in China and Europe are very different. In China, most new factories (65.3%) are "pure-play foundries". This means they are flexible and can quickly make whatever special chips the military or space programs need. In Europe, most new factories (87.2%) are "IDMs" (Integrated Device Manufacturers). These factories only make specific chips for their own products, mostly for cars or everyday industry. Because of this, Europe's space sector must buy its special, low-volume chips from foreign factories. This creates a massive political and security risk for Europe.

3. Re-imagining the Space Supply Chain: What's Next for Europe?

Europe's space supply chains have deep weaknesses. They rely too much on foreign parts and can be easily pressured by other countries. Because of this, Europe needs a completely new way to manage its supply networks. Europe cannot just copy China and force total independence. Its open market and mix of different countries make that impossible. Instead, Europe must fix its weak spots using safe, clear, and data-driven computer systems. These systems can manage a massive, multi-country supply chain with complete trust.

3.1. Blockchain Integration for Provenance and Geopolitical Resilience

Today, building a spacecraft requires thousands of suppliers from all over the world. Right now, it is very hard for big space companies to track exactly where raw materials come from or how tiny micro-components are made. This lack of clear tracking is dangerous. It opens the door for

fake parts, hidden hardware viruses, and sudden supply stops caused by international trade blocks.

To fix these big problems, the space industry can use blockchain technology (also called Distributed Ledger Technology, or DLT). Blockchain offers a mathematically safe way to track space parts from the moment materials are mined to the moment they are launched into orbit. The industry can use a "permissioned consortium blockchain." This means only an approved group—like the ESA, national space agencies, big contractors, and trusted small suppliers—can run the network. This creates total transparency while still keeping company trade secrets safe.

The technical plans for this kind of system are already written in guides like the U.S. NIST Manufacturing Meta-Framework (NIST IR 8536). In a blockchain space system, every important part (like a radiation-hardened memory chip) gets its own unique digital identity, or "digital twin," as soon as it is made.

As the part moves from factory to factory, every step is locked with cryptography and saved forever on the blockchain ledger. This setup brings three massive benefits:

- **Protection against Global Politics:** The system permanently tracks exactly where every part comes from and who owns the company that made it. This means agencies can use computer rules to automatically block parts from dangerous or sanctioned countries. It helps Europe stay independent and guarantees they follow international export rules, like the US EAR and ITAR.
- **Stopping Fake Parts:** The locked blockchain records, combined with special physical security tags (called PUFs) built right into the hardware, mathematically prove that a part is real. This guarantees that no fake, broken, or dangerous parts end up in a flying spacecraft.
- **Real-Time Tracking:** By using a math technique called Zero-Knowledge Proofs (ZKPs), companies can verify supply chain facts without showing their secret prices or business details. This allows big contractors to instantly trust their smaller suppliers.

The European Space Agency (ESA) is already testing this idea. They funded a project called "Track & Trust." This project mixes connected sensors (IoT), satellite communication, and blockchain to track supplies safely, even in tough places with bad internet. In this system, remote sensors collect shipping data. They send it to space using satellites, along with GPS location data. This information is locked onto the blockchain. Anytime someone touches or moves the package, the system automatically checks the math and updates the status.

If the EU expands the "Track & Trust" idea from basic shipping to the actual factory floor, it can create one giant, secure digital network. This would completely remove all the slow paperwork and friction that makes cross-border European space manufacturing so difficult today.

3.2. Future Space Policies: Supranational Consolidation and "Space Team Europe"

To succeed globally, the main advice for the EU is to strongly build the "Space Team Europe" idea. Europe cannot compete against China's highly organized government or the wealthy US private sector if it stays divided. Right now, Europe acts like a loose group of 27 different national space programs.

The European Commission must use the new EU Space Act to force Europe to be technologically independent. Europe has to build its own local factories for vital electronic parts. These include advanced packages, MRAM memories, and power chips. Making these at home will protect Europe from sudden global supply shortages.

Also, the European Space Agency (ESA) needs to change its old "geo-return" rule. This rule gives space contracts to countries based on how much money they donate to the agency. This rule might please politicians, but it slows down business growth and creates a lot of waste.

Instead, Europe should adopt a "buy European, buy bulk" strategy. This means government agencies should promise to buy large amounts of local space products. This steady, guaranteed demand is exactly what small start-ups need. It helps them convince private investors to give them money, allowing them to grow larger and lower their costs.

3.3. Fine-Grain Recommendations for Major Contributors

To make Europe truly independent in space, its biggest countries must play to their specific strengths. Instead of fighting each other for the same business, these nations need to work together. They must align their national plans to build one strong European industry.

France is the leader of Europe's military space defense and the main supporter of the Ariane rocket program. France must learn to balance its strict military needs with the fast pace of private business. First, the French space agency (CNES) needs to speed up the Ariane 6 rocket program. It should also open the Guiana Space Centre (Kourou) to more European start-ups that build small rockets. Turning Kourou into a busy, shared commercial spaceport will guarantee Europe can launch any size of satellite on its own. Second, France has excellent military cyber defense. It should share these strong security rules with the private space market. This will let small French companies safely sell advanced technology originally made for the military, helping the local economy grow.

Germany's biggest strength is its massive manufacturing industry and its research into new materials. The German Aerospace Center (DLR) must work hard to build local European supply chains for advanced power electronics. Germany should build on current projects like EPOSIC and ENSPECT, which focus on new materials for space chips. This will make Germany the undisputed center of Europe for radiation-hardened space chips. By upgrading this technology and building certified factories at home, Germany can stop Europe from relying on American or Asian factories.

Also, Germany supports many private rocket start-ups. To help these young companies survive, the German government must sign long-term contracts to buy their launch services early on.

Italy has a very well-connected space industry. It has a few giant companies and a huge network of small businesses (SMEs) that make up 80 percent of its national space market. The Italian Space Agency (ASI) must use the new Space Economy Law (Law 89/2025) to modernize these small factories rapidly. Italy has a $175 million fund and $8.5 billion from a national recovery plan. It should use this money to digitize its supply chain and connect its small businesses to the new European blockchain networks. Furthermore, Italy is very good at

building pressurized rooms for space. They built big parts of the International Space Station (ISS). Italy should focus its funding on new projects like the Multi-Purpose Habitation (MPH) module. This will give Italy a very important role in future moon missions (the Artemis program) and ensure Europe remains a vital partner for global space exploration.

The UK is no longer in the EU rules framework, but it is still a vital member of the ESA. It must act as a fast, highly specialized partner within the global supply network. In 2026, the UK Space Agency will merge with the DSIT Space Directorate. After this, the UK should stop trying to do everything on its own. Instead, it should focus on dominating a few fast-growing, special niches. The government should only give grant money to areas where the UK is already strong. These areas are tracking objects in space, secure navigation, and fixing things in orbit. Most importantly, the UK is investing heavily in Active Debris Removal (ADR) to clean up space junk. By mastering how to fix satellites and clean up trash, the UK can set the global rules for keeping space clean. The UK can then sell these highly required services to European and American satellite companies, giving the UK massive political power.

4. Conclusion

The global space economy is changing fast. Today, countries are using supply chains as weapons, the commercial market is growing rapidly, and nations absolutely must control their own technology. China is a perfect example of this. They build everything themselves, from their space station to their older computer chips. Because of this, the old way of buying space parts from all over the world is no longer safe.

If Europe wants to be a powerful leader in tomorrow's $1.8 trillion space market, it must completely change its strategy. European countries can no longer just focus on their own separate goals. Instead, they must work together to build one strong, united industry, guided by new rules like the EU Space Act.

Most importantly, Europe must fix its weak physical supply chains by using its great strengths in software and law. By using blockchain technology, Europe can mathematically track every single part and guarantee it is safe and real. This will create a highly secure manufacturing network across many countries. Ultimately, to protect its massive future in space, Europe must first secure the tiny microchips at the very bottom of its supply chain.

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